Structural basis of ZH853 recognition by the µ-opioid receptor: a distinctive extracellular-loop contact defines a selectivity handle and guides analog design

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Status: Results for Objectives 1–3 are complete; the molecular-dynamics and free-energy validation (Methods §2.5) is prepared but not yet run — claims dependent on it are flagged [**prospective**].

Abstract

The cyclic endomorphin-1 analog ZH853 is a potent µ-opioid receptor (MOR) agonist with an attractive preclinical profile, but the structural basis of its binding — and how it differs from other opioid agonists — has not been characterized. Using a 3.5 Å cryo-EM structure of the MOR–ZH853–Gi–scFv16 complex, we computed heavy-atom interaction fingerprints for ZH853 and benchmarked them against thirteen deposited MOR complexes spanning peptide agonists (DAMGO, endomorphin-1, β-endorphin), morphinan and fentanyl-class small molecules, biased agonists, and an antagonist. ZH853 engages the canonical opioid anchor (a salt bridge to Asp3.32 satisfied by its Tyr1 α-amine, plus the Tyr7.43 "3–7 lock") shared by all agonists, but additionally forms a **salt bridge to Glu231 in extracellular loop 2 (ECL2) that no other agonist in the set makes**, together with an aromatic/cation-π contact to His7.36 shared only with its endomorphin-1 parent. This extended ECL2/ECL3 engagement is the structural signature of the cyclic scaffold. We nominate **E231Q/E231A** as mutations predicted to disrupt ZH853 selectively while sparing related full agonists, and identify the molecule's beyond-rule-of-5 liabilities (TPSA 235–280 Å^2, 8–10 H-bond donors), proposing two orthogonal optimization series — backbone N-methylation for passive permeability and semaglutide-style C-terminal lipidation for half-life — targeted to solvent-exposed positions established from the bound pose. A membrane molecular-dynamics and free-energy protocol to test these hypotheses is established and released. This is, to our knowledge, the first structural analysis of a cyclic-peptide MOR complex.

1 Introduction

Opioid analgesics acting at the µ-opioid receptor (MOR, gene *OPRM1*) remain the mainstay of severe pain management, but their utility is limited by respiratory depression, tolerance, and dependence. A central goal of contemporary opioid pharmacology is to retain analgesic efficacy while separating it from these liabilities — through signaling bias, partial agonism, or novel chemotypes. Endomorphins, the endogenous MOR-selective tetrapeptides (Tyr-Pro-Trp-Phe-NH_2), are attractive scaffolds because of their high selectivity, but their linear form is metabolically labile and poorly bioavailable. The Zadina laboratory addressed this by engineering cyclic, D-amino-acid endomorphin analogs; **ZH853** (Tyr-cyclo[D-Lys-Trp-Phe-Glu]-Gly-NH_2) is a lead from this series with potent antinociception and a reduced side-effect profile in rodents [1].

Despite this promise, no structure of ZH853 — or of any cyclic-peptide MOR complex — has been reported, leaving open three questions this work addresses computationally from a 3.5 Å cryo-EM structure of the MOR–ZH853–Gi–scFv16 complex: (1) *what, if anything, is structurally distinctive about how ZH853 engages MOR* relative to other agonists; (2) *which receptor mutations would abrogate ZH853 while sparing related full agonists*, providing a pharmacological fingerprint and a route to mechanistic tests; and (3) *how ZH853's drug-like properties could be improved*, drawing on modern peptide-drug strategies. We further establish and release a membrane molecular-dynamics (MD) and free-energy protocol to validate these structure-derived hypotheses.

2 Related work

MOR structural pharmacology. The inactive MOR structure (β-FNA, 4DKL [2]) and the active BU72 complex (5C1M [3]) established the orthosteric pocket and the activation-associated rearrangements of the W6.48 toggle and TM6. Cryo-EM of MOR–Gi complexes then resolved a series of agonists — the peptide DAMGO (6DDE [4]; human 8EFQ) and endomorphin-1/β-endorphin (8F7R/8F7Q [7]), and the small molecules fentanyl, morphine, oliceridine, SR-17018, PZM21 (8EF5/8EF6/8EFB/8EFL/8EFO [5]) and mitragynine pseudoindoxyl/lofentanil (7T2G/7T2H [6]). All share the D3.32 anchor and a conserved hydrophobic subpocket; biased agonists differ mainly in TM6/ECL2 engagement. Notably, *every* deposited peptide-agonist MOR structure carries a *linear* peptide — ZH853 would be the first cyclic-peptide complex.

Cyclic peptides as drugs. Macrocyclic peptides occupy "beyond-rule-of-5" (bRo5) chemical space [9]; their oral/passive permeability is governed less by static polarity than by conformational shielding of backbone donors ("molecular chameleons" [10]), which N-methylation and cyclization promote (as in cyclosporine). Half-life, by contrast, is engineered through albumin-binding **lipidation** — the strategy that gives semaglutide a $\sim$1-week half-life via a γGlu-2$\times$AEEA-C18-diacid conjugate [11]. These two axes — permeability and duration — are distinct and are treated separately here.

3 Methods

3.1 Structure and numbering. We used the deposited real-space-refined MOR–Gi–scFv16–ZH853 model (3.5 Å; CC_{mask} 0.80; 0.0% Ramachandran outliers). Chains: Gαi (A), Gβ1 (B), Gγ2 (C), scFv16 (D), MOR/*OPRM1* residues 69–349 with 84 modeled cholesterols (R), and ZH853 as HETATM ligand L01 (E). The construct uses **human OPRM1 (UniProt P35372) numbering**, verified against all canonical orthosteric positions; mouse-numbered comparators were offset by +2 to this frame. The ligand's 59 heavy atoms and formula ($C_{42}H_{51}N_9O_8$, 810 Da) match the reference SMILES, confirming the monomeric macrocycle.

3.2 Comparator set. Thirteen structures were retrieved from the RCSB PDB: peptide agonists 6DDE/8EFQ (DAMGO), 8F7R (endomorphin-1), 8F7Q (β-endorphin), 9WST/9WSV (DAMGO with Gz/β-arrestin); small molecules 5C1M (BU72), 8EF5 (fentanyl), 8EFB (oliceridine), 8EFL (SR-17018), 8EFO (PZM21), 7T2G (mitragynine pseudoindoxyl); and antagonist 4DKL (β-FNA). For each, the receptor chain and bound agonist were isolated programmatically and mapped to human OPRM1 numbering.

3.3 Interaction fingerprints. Because the 3.5 Å model lacks hydrogens, we used a resolution-matched **heavy-atom geometric fingerprint** (rather than angle-based H-bond criteria), classifying each receptor–ligand residue contact as ionic (opposite-charge groups $\leq$4.0 Å), H-bond (polar N/O $\leq$3.5 Å), hydrophobic (C–C $\leq$4.0 Å), π-stacking (aromatic centroids $\leq$5.5 Å), or cation-

π ($\leq$6.0 Å). Aromatic rings of non-standard ligands were detected by planar-ring geometry, a template-free method that generalizes across chemotypes. The same method was applied uniformly to all fourteen complexes.

3.4 Mutation panel and cheminformatics. Candidate selectivity mutations were scored by ZH853 interaction strength $\times$ rarity across the twelve agonists $\times$ a bonus for sparing the two closest full agonists (endomorphin-1, DAMGO). Physicochemical descriptors (2D, plus 3D radius of gyration, polar SASA fraction, and intramolecular H-bond count from macrocycle-aware conformers) were computed with RDKit; modification variants (N-methylation, lipid conjugation, halogenation) were enumerated programmatically and their properties predicted. Buried vs solvent-exposed ligand atoms were determined from the bound pose ($\leq$4.5 Å to receptor).

3.5 MD/free-energy protocol [prospective]. A membrane MD system was prepared: receptor sidechains rebuilt (PDBFixer), protonation assigned by PROPKA at pH 7.4, and ZH853 protonated to +1 and prepared for GAFF2/RESP parameterization. PROPKA places **D2.50 (Asp116) at pKa 7.61**, so parallel protonated/charged systems are built. The production protocol (POPC:cholesterol 9:1, ff19SB/Lipid21/OPC, CHARMM-GUI-style six-stage equilibration, LangevinMiddle 310 K with a semi-isotropic membrane barostat, hydrogen-mass repartitioning at 4 fs, $\geq$3 replicas) and free-energy calculations (relative FEP across the ZH850/831/809 analogs; ABFE and funnel metadynamics as exploratory) are released as a SLURM bundle. Given documented convergence difficulties for flexible macrocycles, free-energy results are framed as rank-ordering.

All analyses are scripted and version-controlled (Data & code availability).

4 Results

4.1 ZH853 satisfies the conserved opioid anchor

ZH853 contacts 25 receptor residues. Its Tyr1 α-amine forms the **canonical Asp3.32 (Asp149) salt bridge** (3.16 Å) and hydrogen-bonds Tyr7.43 (Tyr328, 3.17 Å) — the "3–7 lock" — while the Tyr1 ring and the macrocyclic Trp/Phe pack into the conserved hydrophobic subpocket (Met3.36, Val3.28/Ile3.29, Ile6.51, Val6.55, Val5.42). Every one of these contacts is shared by all twelve comparator agonists (Figure 1), confirming that ZH853 presents the same "message" pharmacophore as morphine, fentanyl, DAMGO, and endomorphin-1. Consistent with this, PROPKA assigns Asp149 a depressed pKa (6.0), i.e. charged and salt-bridge-competent.

4.2 A distinctive ECL2 salt bridge distinguishes ZH853 (Objective 1)

Against this conserved background, three contacts set ZH853 apart (Figure 1; Table 1):

- **Glu231 (ECL2): an ionic + H-bond contact made by ZH853 alone (0/12 agonists).** A ZH853 basic group reaches ECL2 to form a salt bridge (3.12 Å) that no linear peptide or small molecule in the set reproduces. PROPKA confirms Glu231 is charged (pKa 4.9). This is the single strongest structural discriminator.
- **His7.36 (His321): aromatic/cation-π, shared only with endomorphin-1 and mitragynine (2/12).** An ECL3-proximal contact retained from the endomorphin-1 lineage.
- Weaker ZH853-unique contacts at Leu221/Phe223 (ECL2) and Leu234 (TM5), gained by the extended/cyclic scaffold.

The picture is coherent: **ZH853 keeps the deep conserved anchor but reaches "up and out" toward ECL2/ECL3**, engaging a rim region that the compact linear agonists do not. This extended loop engagement is the structural hallmark of the macrocycle.

4.3 A selectivity handle: E231 mutations (Objective 2)

Ranking pocket residues by ZH853 interaction strength, rarity across agonists, and sparing of the endomorphin-1/DAMGO parents nominates **Glu231 as the top selective target** (Table 1). We predict **E231Q** (charge-neutral isostere) and **E231A** (removal) will attenuate ZH853 binding while leaving agonists that do not contact ECL2 Glu231 largely unaffected — a testable pharmacological signature. His7.36 substitutions (H321A/F) are a secondary, less clean handle (they also perturb endomorphin-1). By contrast, the universal anchors (Asp3.32, Tyr7.43, Tyr3.33, Met3.36, and the hydrophobic cage) are **do-not-mutate controls**: disrupting them abrogates all agonists and cannot confer selectivity. [**prospective**] magnitudes await MD occupancy and relative-FEP/in-silico-mutagenesis validation.

4.4 Drug-likeness liabilities and a two-axis design strategy (Objective 3)

All four analogs sit firmly beyond rule-of-5 (MW 714–810, TPSA 235–280 Å^2, 8–10 H-bond donors; Figure 2); the dominant liabilities are high polar surface area and donor count. Their 3D conformers show 4–6 intramolecular H-bonds — partial donor self-shielding consistent with chameleonic behavior. From the bound pose, **46 of 59 ligand heavy atoms are buried** (the Tyr1/aromatic pharmacophore) and **13 are solvent-exposed**, defining where derivatization is structurally safe. We therefore propose two orthogonal series (Figure 3):

1. **Permeability** — backbone **N-methylation** of solvent-facing amides (each removes one donor and $\sim 9\,\text{Å}^2$ TPSA; a 2–3-site subset reaches TPSA $\approx 253\,\text{Å}^2$, crossing the oral-bRo5 threshold), optionally with 4-F-Phe for metabolic stability. Best pursued from **ZH831**, the least liable analog.

2. **Half-life** — **C-terminal semaglutide-style lipidation** on the exposed cap (γGlu-2×AEEA-C18-diacid), trading polarity for albumin-mediated duration; pursued from the most potent analog, **ZH853**.

Both series must be potency-checked against MOR by relative FEP, and N-methyl sites chosen to avoid the receptor-facing backbone amides identified in §4.1–4.2.

5 Discussion

The central finding is that ZH853's distinctiveness lies not in the conserved orthosteric anchor — which it shares with all opioid agonists — but in a **cyclic-scaffold-enabled ECL2 salt bridge (Glu231)** and adjacent ECL3 aromatic contact (His7.36). ECL2 is a recognized determinant of ligand selectivity and kinetics across GPCRs, and its engagement here provides a plausible structural correlate for ZH853's distinct pharmacology and a concrete, low-ambiguity selectivity handle (E231Q). Because Glu231 is peripheral to the conserved message, its mutation is well suited to a clean pharmacological separation of ZH853 from morphinan/fentanyl chemotypes and even from its linear endomorphin-1 parent.

The design analysis reframes ZH853 optimization as two independent problems. The molecule already banks the hard-won macrocyclic and D-amino-acid protease resistance; what remains is permeability (a donor/PSA problem, addressable by N-methylation guided by which amides face solvent vs receptor) and duration (a half-life problem, addressable by lipidation on the exposed cap). Coupling the design to the bound pose avoids the common error of derivatizing the pharmacophore.

Limitations. All results derive from a single 3.5 Å static model; sidechain rotamers and the exact ligand pose carry real uncertainty, and interaction assignments use heavy-atom geometry rather than explicit hydrogens. The comparative analysis is a contact census, not an energetic ranking. The mutation and design predictions are hypotheses; their magnitudes require the MD/FEP validation whose protocol we establish here, and macrocycle free-energy convergence is itself a known challenge.

6 Conclusions and future work

From the first structural analysis of a cyclic-peptide MOR complex, we identify an ECL2 Glu231 salt bridge as the defining, ZH853-specific interaction, nominate E231Q/E231A as selectivity-conferring mutations, and propose bound-pose-guided N-methylation and lipidation series to address the molecule's permeability and half-life liabilities. The immediate next step is to run the released membrane-MD and free-energy protocol to (i) confirm contact occupancy and pose stability, (ii) quantify the E231 selectivity, and (iii) rank the analog and designed-variant affinities. Longer term, the Gz and β-arrestin comparators included here support a follow-on analysis of whether ECL2/ECL3 engagement correlates with ZH853's reported signaling bias.

Figures

Table 1: ZH853 contacts ranked by distinctiveness (excerpt). Full interaction matrix and mutation panel are provided as dated `product/` outputs (Data and code availability).

Residue	BW	ZH853 interaction	#/12	Note
Glu231	ECL2	ionic + H-bond	**0**	ZH853-unique; primary selectivity handle (E231Q/E231A)
His321	7.36	π-stack + cation-π	2	shared only with endomorphin-1, mitragynine
Tyr130	2.64	hydrophobic	1	endomorphin-1 only
Leu221/Phe223	45.52/45.54	contact	0	weak ECL2 contacts (cyclic scaffold)
Asp149	3.32	ionic + H-bond	12	universal anchor — do-not-mutate control
Tyr328	7.43	H-bond	12	universal (3–7 lock) — control

Data and code availability

All analyses are scripted and version-controlled in the project repository (`src/`, numbered workflows; `zh853mor` package; `make analysis` reproduces every figure and table). Comparator structures are re-fetchable via `make fetch`. Decisions and clarifications are logged in `SPECIFICATION.md`; detailed methods and results are in the `docs/` directory (`METHODS_md_prep`, `RESULTS_interactions`, `RESULTS_analog_design`).

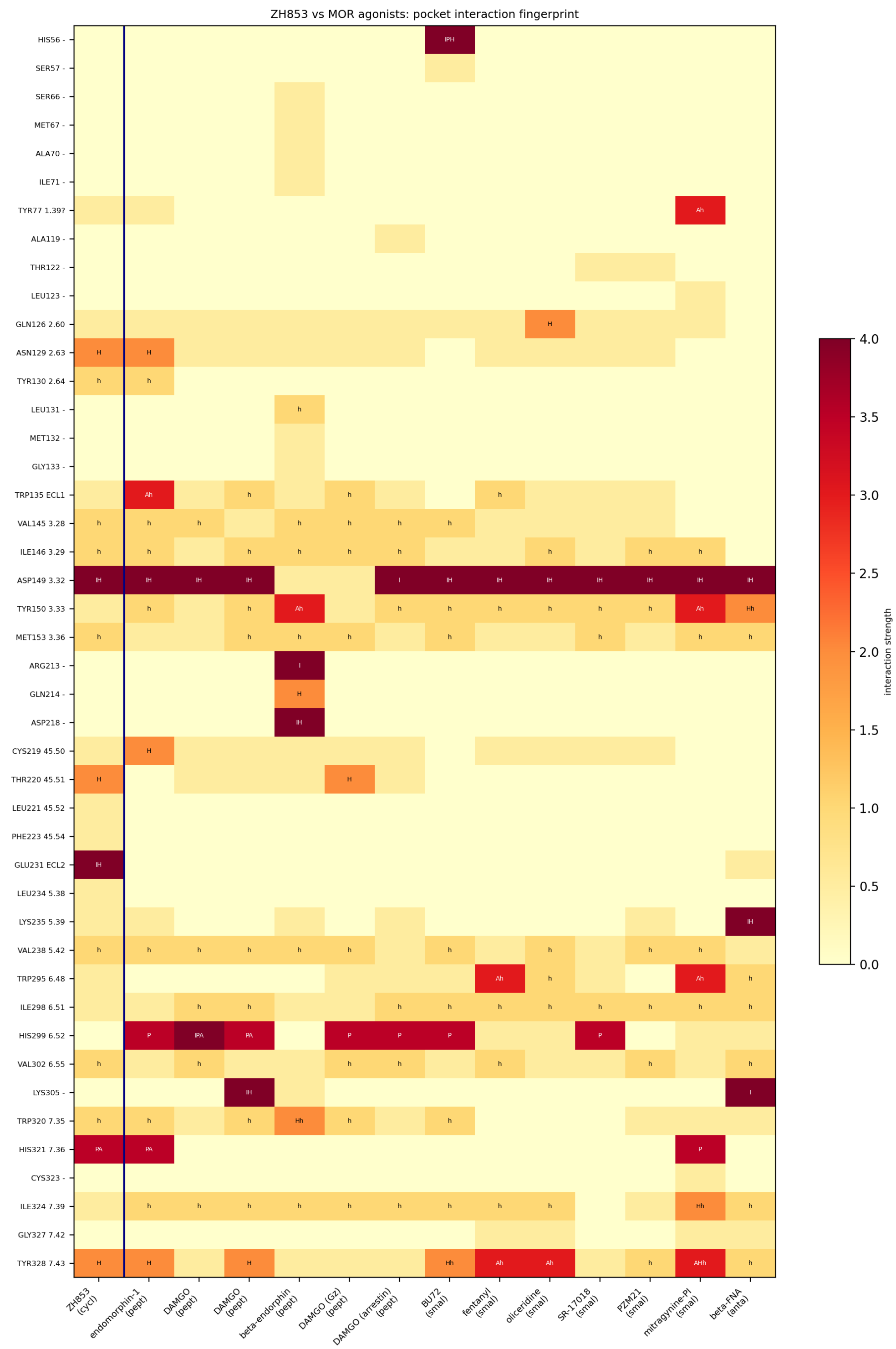

Figure 1: Pocket interaction fingerprint of ZH853 vs 13 MOR complexes. Rows = receptor residues (human OPRM1 / Ballesteros-Weinstein); columns = complexes; cells colored by interaction strength and labeled by type (I ionic, P cation-π, A π-stack, H H-bond, h hydrophobic). Asp3.32 is engaged across all columns; Glu231 (ECL2) is engaged in the ZH853 column alone.

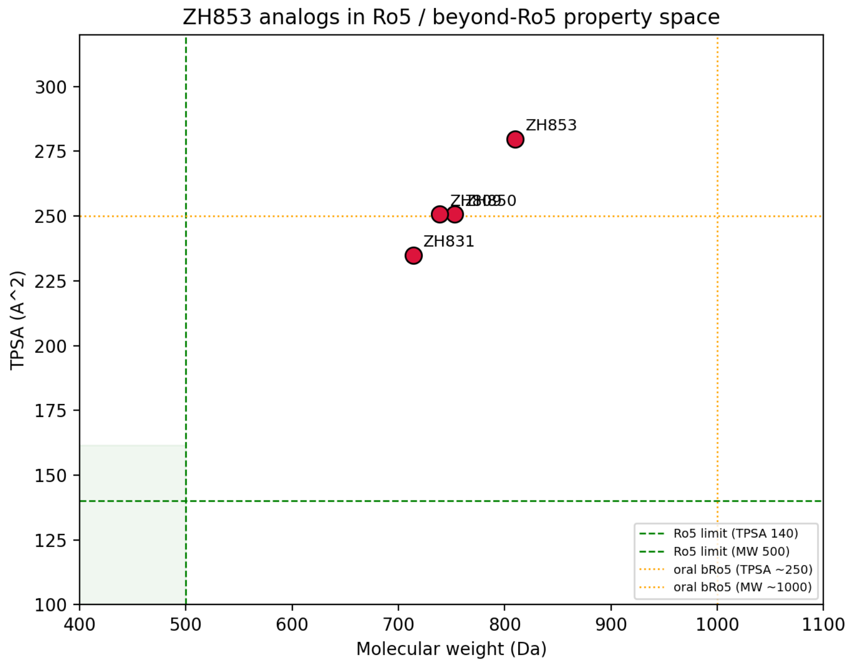

Figure 2: ZH853 analogs in Ro5 / beyond-Ro5 property space (MW vs TPSA), with Lipinski and oral-bRo5 reference limits.

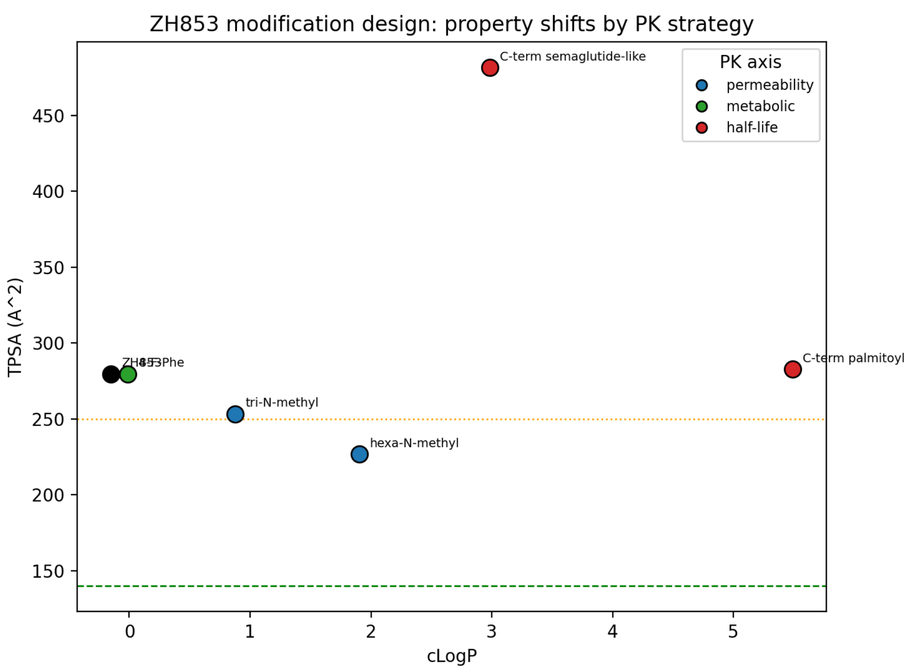

Figure 3: Predicted property shifts of the modification strategies by PK axis (TPSA vs cLogP): N-methylation (permeability) lowers TPSA across the bRo5 line; lipidation (half-life) raises lipophilicity.

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